Which data improve final-grade predictions?

Bayesian DAG, posterior simulation, RETRO text, model comparisons, grading weights and summer exclusion.

Research question

What do the inputs add to predictions—and how uncertain are those conclusions?

Current conclusion

Keep the simple grade-based baseline.

One current account of completed work; later improvements are labelled as proposals.

191 students, six non-summer terms

Fall 2024–Spring 2026; one analysis row per student. SMR/Summer courses are excluded.

Grades

Block A: 10 scores
Block B: 16 scores

LMS

Log page views
Log participations

EVC + wellness

Three EVC-style inputs
One wellness mean

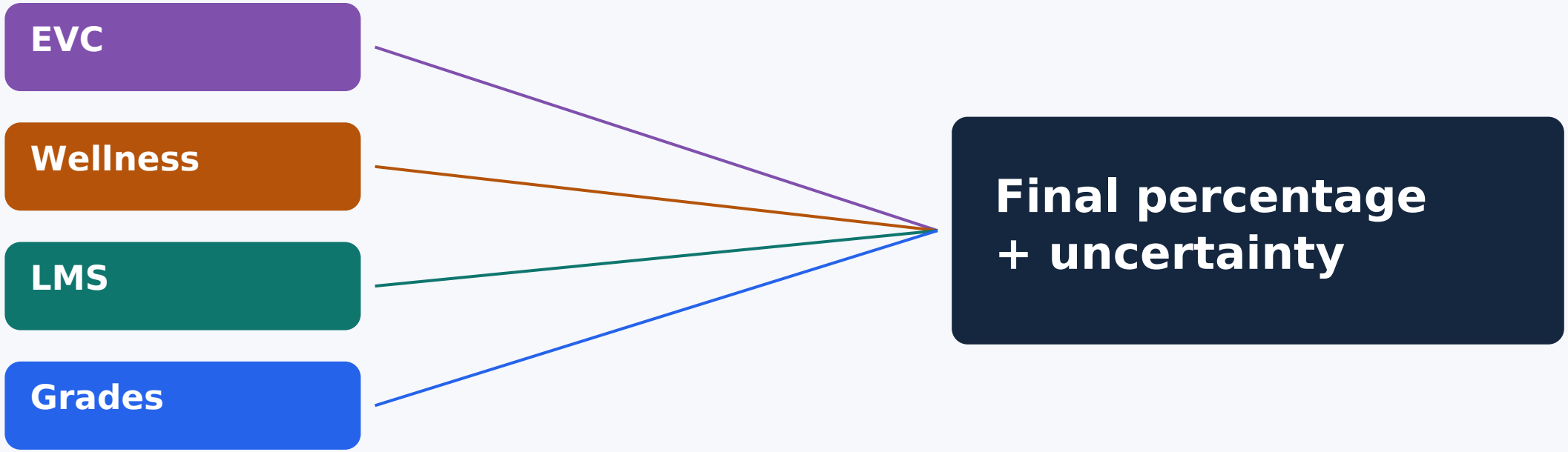
RETRO text

Three open prompts
Three covered terms

78/191 lack early surveys. Exact grade-release and RETRO submission times remain unverified.

The DAG makes assumptions visible

The original fixed graph has 23 nodes and 62 arrows; its structure was not learned from correlations.



Outcome-parent view only. Arrows are assumptions; causal and coaching effects are not established.

Three questions need different evidence

Posterior signs, individual outcome risk and predictive improvement are separate quantities.

Association

How plausible is a positive or practically large conditional weight?

Outcome

What probability does the model assign to a final result below 70%?

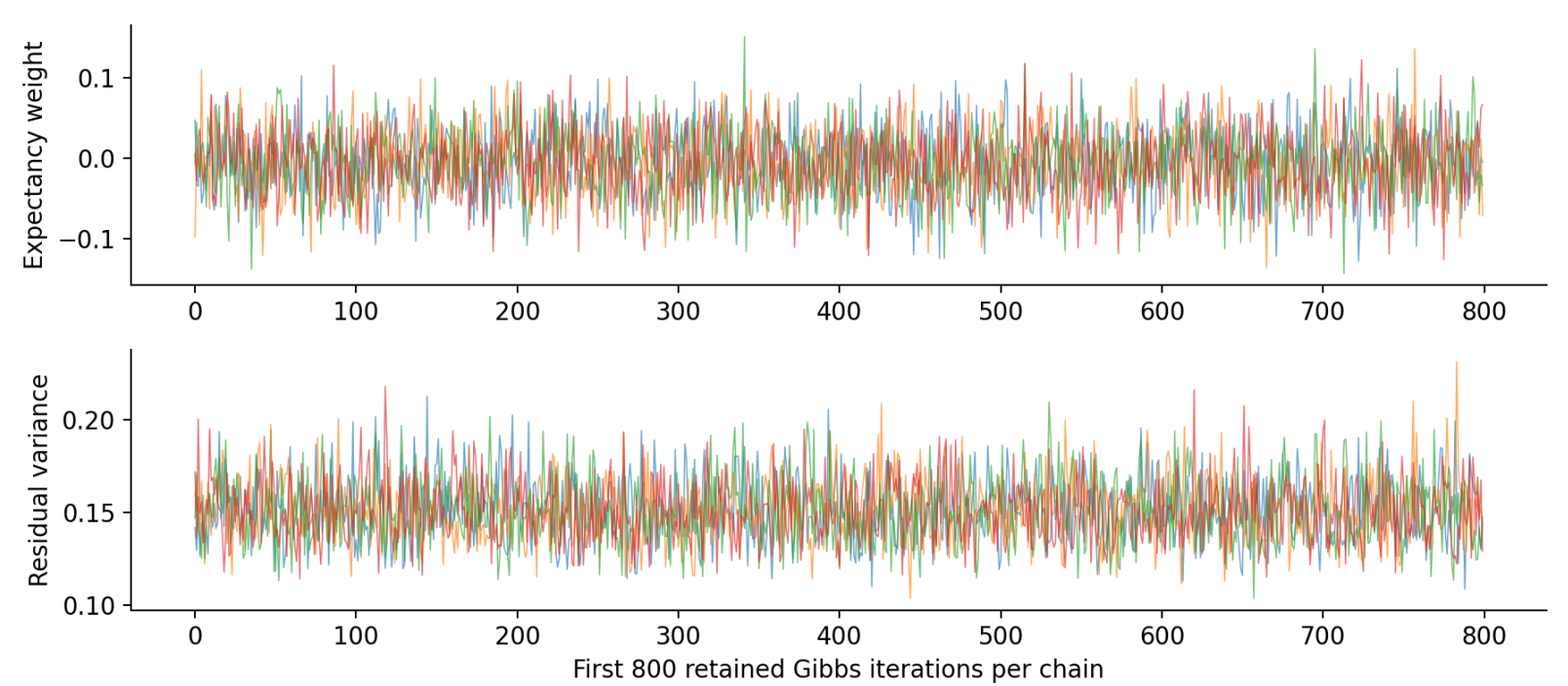
Added value

Does the source reduce error for the same later students?

Changing an input in a prediction is not evidence that intervening on it changes a grade.

Posterior simulation agrees with the exact solution

20,000 independent posterior draws; four actual Gibbs MCMC chains.



Numerical checks

Max R-hat: 1.0007
Min bulk ESS: 12,194
Means within 2.01 MCSE

Outcome-CPD sampling conditional on the imputed design—not full missing-data DAG inference.

Learn from earlier terms; check later students

Four expanding-window evaluations; 146 later-student predictions per numeric model.

Train through	Evaluate	Students
Winter 2025	Spring 2025	43
Spring 2025	Fall 2025	20
Fall 2025	Winter 2026	40
Winter 2026	Spring 2026	43

Compare: no grades / Block A / Block B. Main comparisons and all sensitivities are retained.

Historically explored data and settings; course timing is not verified.

All-source additions did not improve grade-based MAE

Macro MAE gives each evaluation term equal weight. Positive reduction means lower error.

Grade evidence	Baseline	All numeric sources	MAE reduction
No grades	6.538	6.524	+0.014
Block A	3.607	3.695	-0.089
Block B	2.948	3.036	-0.088

Block B combined model worsened MAE in all four later terms.

LMS has a modest signal when grades are absent

Exploratory source-specific comparison; a no-grades scenario is not itself a verified early forecast.

Inputs beyond mean baseline	MAE	MAE reduction	Terms better
LMS activity	6.36	+0.177	4/4
Wellness	6.56	-0.020	2/4
EVC	6.71	-0.173	0/4
LMS + wellness	6.34	+0.198	4/4
All numeric sources	6.52	+0.014	2/4

The baseline is the earlier-student mean; its macro MAE is 6.54 points.

Conditional effects are small and uncertain

Final fit: 148 earlier students. Final-percentage points per one training standard deviation.

Input	Mean points / SD	95% credible interval
LMS page views	+0.34	[-0.21, +0.88]
LMS participations	-0.09	[-0.63, +0.44]
EVC source cost/support	-0.21	[-0.84, +0.41]
EVC expectancy	-0.05	[-0.70, +0.60]
EVC value	-0.13	[-0.73, +0.47]
Wellness	+0.20	[-0.39, +0.78]

All six P(|conditional weight| > 1 point) values are below 1%, conditional on this model.

The new RETRO source contains open reflections

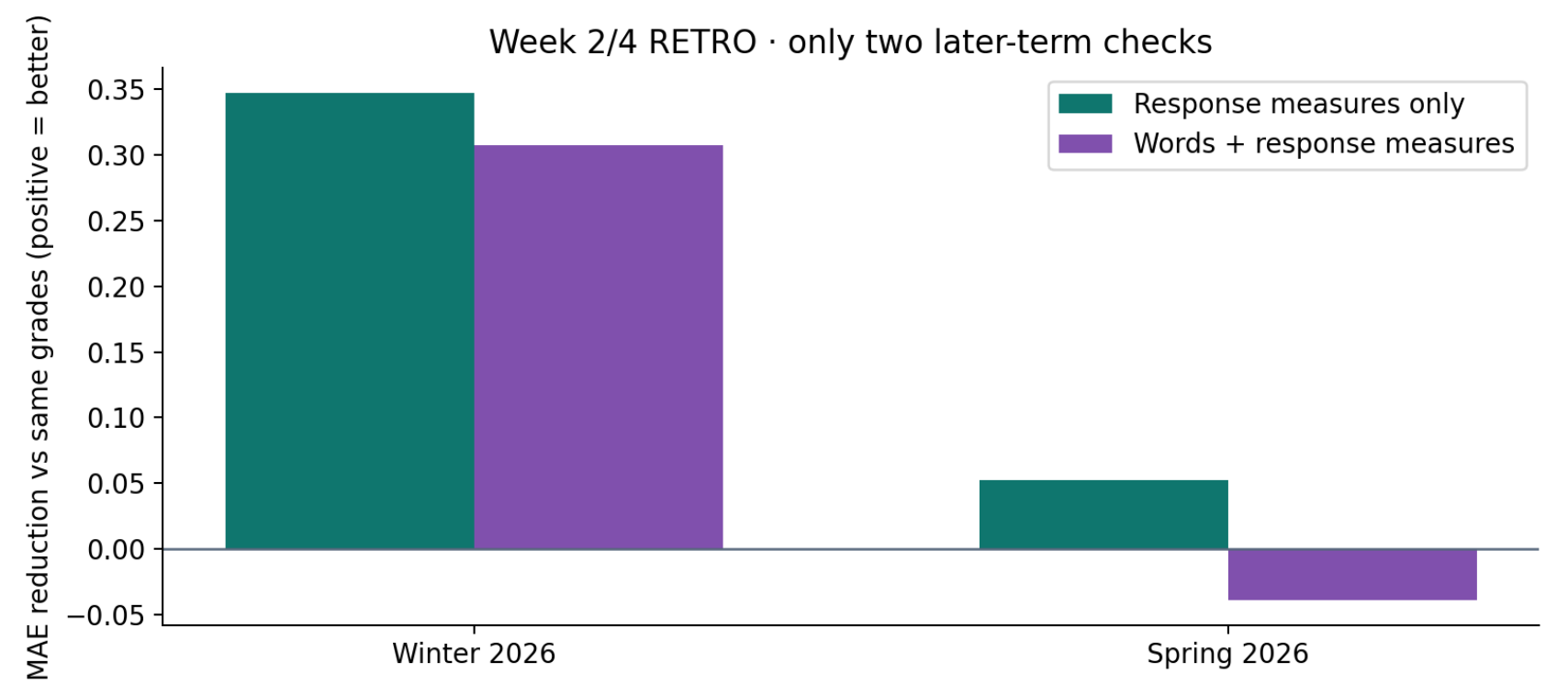
870 response rows; 74 students; Weeks 2, 4, 6, 8 and 10; all IDs match the existing records.

RETRO term	Grade-table students	Week 2/4 respondents
Fall 2025	20	13
Winter 2026	40	25
Spring 2026	43	30

Only week labels are available. Earlier terms are not coded as nonrespondents.

RETRO words show mixed added value

Train-only TF-IDF plus response measures; larger grades-only baseline; Ridge alpha 10.



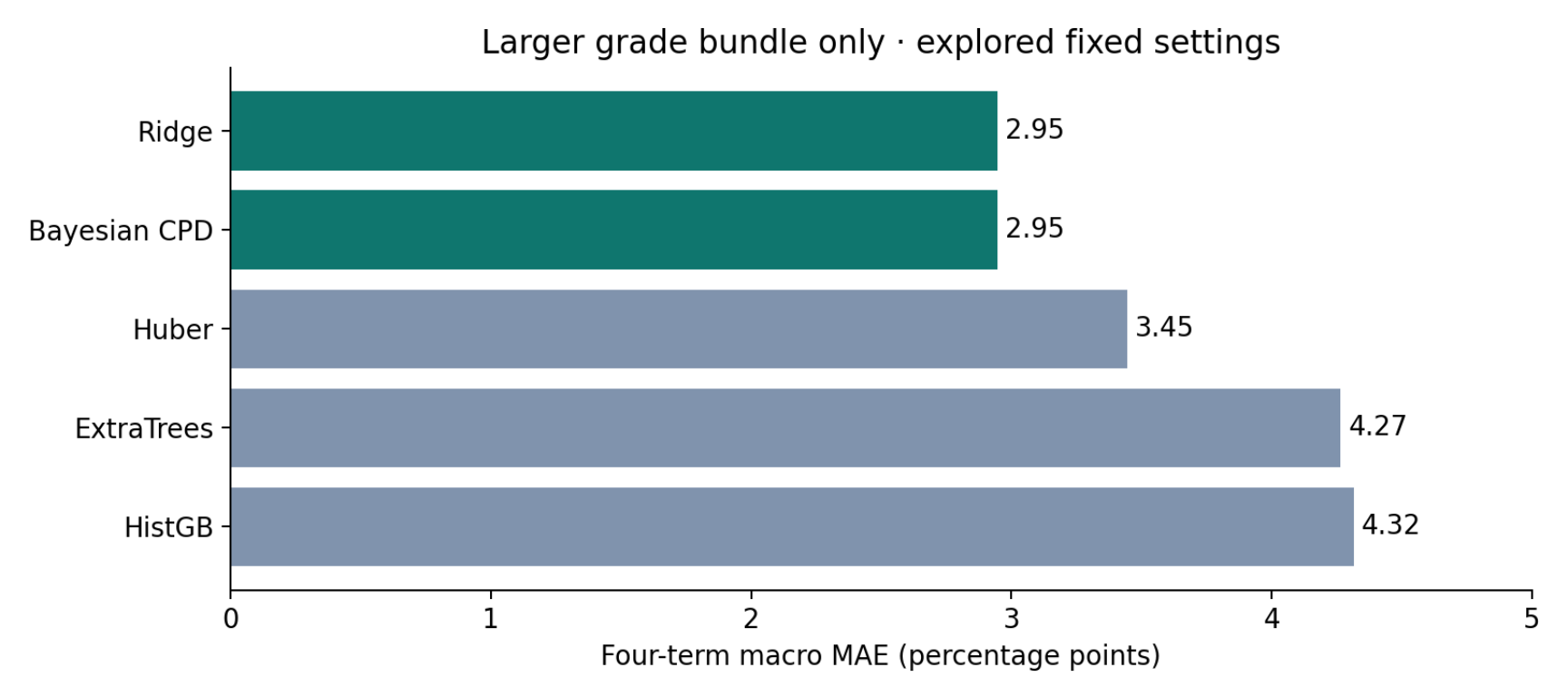
Interpretation

No consistent independent content benefit established.

Two explored evaluation terms; no claim of an identified causal RETRO effect.

Tested alternatives did not beat the grades baseline

Same four numeric temporal checks; fixed, modest candidate settings; point error only.



Decision

Keep the simple baseline while improving timing and measurement.

Posterior probabilities can still miss low outcomes

Illustrative below-70 threshold; it is not an official pass/fail or intervention rule.

Complete-input example

Mean: 71.24%
90% interval: 65.72–76.75%
P(final < 70): 35.6%
Actual: 75.95%

A low result the model missed

Mean: 77.60%
90% interval: 71.36–83.83%
P(final < 70): 2.3%
Actual: 68.03%

Missing-input uncertainty and unusual failures remain important model weaknesses.

Confirmed weights and explicit summer exclusion

The user confirmed the same main category weights for all six historical non-summer terms.

Main grading category	Course weight
Written Mastery I	20%
Written Mastery II	35%
Verbal Mastery I	10%
Verbal Mastery II	10%
Teamwork Mastery	10%
Life-Long Learning Mastery: Growth Mindset	10%
Life-Long Learning Mastery: Resilience & Agency	5%

Extra credit: Surveys +0.75 points; Team Manager +0.25 points; MLC 0 points.

Predict remaining performance after earned marks

Accounting is implemented and tested; real historical remaining-work targets are not yet reconstructed.

Final total = known earned contribution
+ (1 − known course weight) × remaining main-work percentage
+ extra credit

Category weights are confirmed. Effective historical assignments and availability still need verification.

Required next input: original complete grade exports/source ZIP, extra credit, and release/revision history.

Improve timing and measurement before complexity

A proposed longitudinal design—not a newly fitted or causally identified graph.

Earlier marks

Observed at time t

→

Later EVC / RETRO

Measurements at time t+1

→

Future coursework

Activity / marks at t+2

1 Verify a useful prediction date and remaining-work target.
2 Validate survey scoring, RETRO themes and response patterns.
3 Test restrained term pooling, robust residuals and missing-data models.

Technical reference: exact Bayesian regression

Normal–Inverse-Gamma outcome CPD, conditional on training-only scaling and a fixed imputed design.

$z = A\beta + \varepsilon;\ \varepsilon \sim \text{Normal}(0, \sigma^2 I)$

$\beta \mid \sigma^2 \sim \text{Normal}(0, \sigma^2 \Lambda_0^{-1});\ \sigma^2 \sim \text{InvGamma}(2,1)$

$L = \Lambda_0 + A^T A;\ m = L^{-1} A^T z;\ a = 2 + n/2$

$b = 1 + \tfrac{1}{2}(z^T z - m^T L m)$

Direct: $\sigma^2 \sim \text{IG}(a,b);\ \beta \mid \sigma^2 \sim \text{Normal}(m, \sigma^2 L^{-1})$

Gibbs variance: $\text{IG}(a + p/2,\ b + \tfrac{1}{2}(\beta - m)^T L (\beta - m))$

Predictive: Student-t, df=2a, location $A^* m$, scale$^2 = (b/a)(1 + A^* L^{-1} A^{*T})$

Predictive Student-t from a Gaussian likelihood is distinct from a robust Student-t likelihood.

Technical reference: sensitivity and score limits

All-source additions to Block B remain negative under the recorded sensitivity checks.

Sensitivity	Macro MAE reduction
Main: precision 10	−0.088
Precision 1	-0.191
Precision 100	-0.007
Missingness indicators	-0.119
Complete non-grade respondents	-0.106

Term bootstrap ≠ coefficient posterior ≠ a calibrated probability of success in a new term.

Technical reference: evidence and reproducibility

The organized package contains current deliverables, results, code and input requirements.

01_Presentation/ Consolidated deck, PDF, speaker notes
02_Report/ Current research report and artifact index
03_Project/ Four notebooks, code, tests and all result folders
04_Diagrams/ Full original DAG + labelled temporal proposal
05_Archive/ Earlier decks, reports and supplied reference material

14 tests pass. Remaining-work arithmetic is tested; a historical remaining-work model is not fitted. Historical release/revision snapshots remain missing.

Method references: Murphy (2012), Bayesian linear regression.
Rubin (1981), Bayesian Bootstrap, DOI 10.1214/aos/1176345338.
Vehtari et al. (2021), MCMC diagnostics, DOI 10.1214/20-BA1221.